SUBJ: Miami Center SOP

This document serves as an in depth look into the Miami (ZMA & ZMO) Center operations. The airspace is broken up into different areas, containing different sectors, with descriptions and details of each sector in the pages that follow.

In all cases where FL180 appears in this document, the lowest usable flight level shall be used in lieu of FL180 when it is not usable due to low altimeters.

Exceptional and/or unusual situations may require a deviation or supplementary procedure from this SOP. Use good judgment to resolve this while providing exceptional service.

In all cases where a strikethrough is used in this document, it shall be interpreted as all transitions of the specified procedure *excluding* the one stricken. For example, "~~OMN~~ [**HILEY#** / ANNEY#]**"** refers to all transitions on the latest versions of the HILEY/ANNEY arrivals except for the OMN transition. RNAV routings are indicated by bold text, and are preferred over non-RNAV for capable aircraft. Where SOP routings are enclosed in angle brackets, such as in “>Q77 SHRKS< LAIRI LARZZ”, any routing outside the angle brackets need not be enforced.

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1. PURPOSE

This order prescribes local air traffic control procedures and phraseology for use by Miami Air Route Traffic Control Center personnel on the VATSIM Network. Familiarity and compliance with all parts of this directive is mandatory for all ZMA controllers operating any Miami Center sector.

2. CANCELLATION

This document replaces and cancels ZMA Directive 2, version D, dated 06/04/2022.

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3. AIRSPACE OVERVIEW

Miami Center airspace contains "domestic" and "oceanic" portions. In both areas, the overwhelming majority of traffic receives radar separation because radar coverage is generally good. In the remote areas in the Caribbean, radar coverage can be poor at low altitude, requiring application of non-radar separation. Please note that a controller using VRC will have "perfect" radar coverage, while a controller with properly configured vERAM will not.

The “Florida Centerline” is the area highlighted in orange in Figure 3. The line splits the state into east and west sides, and should be used to determine correct altitudes in accordance with Section 4.a on page 6.

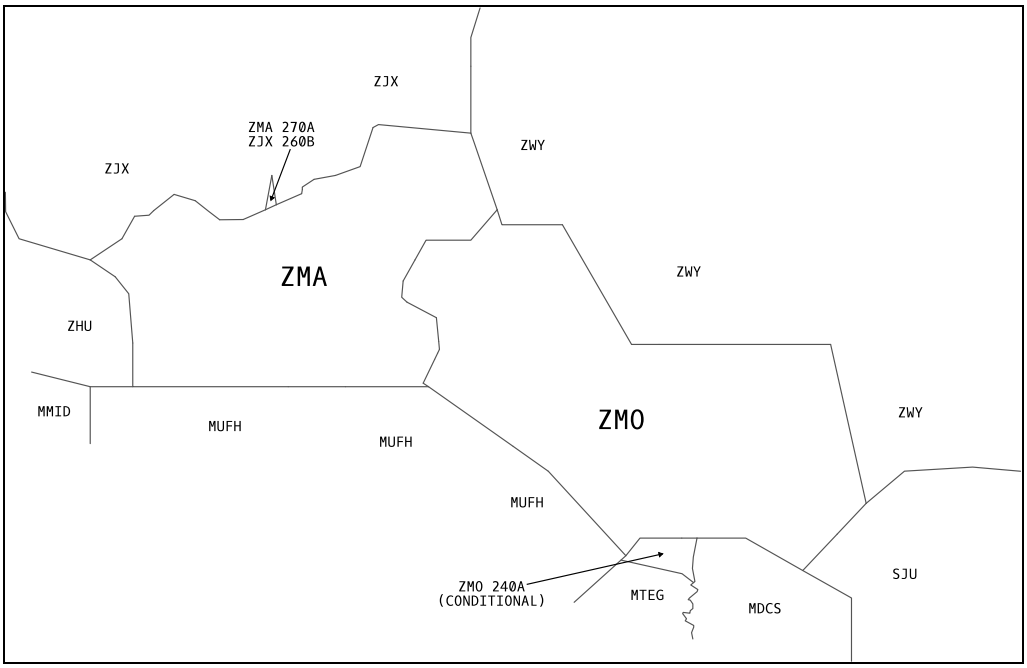

Figure 1 – ZMA/ZMO Airspace

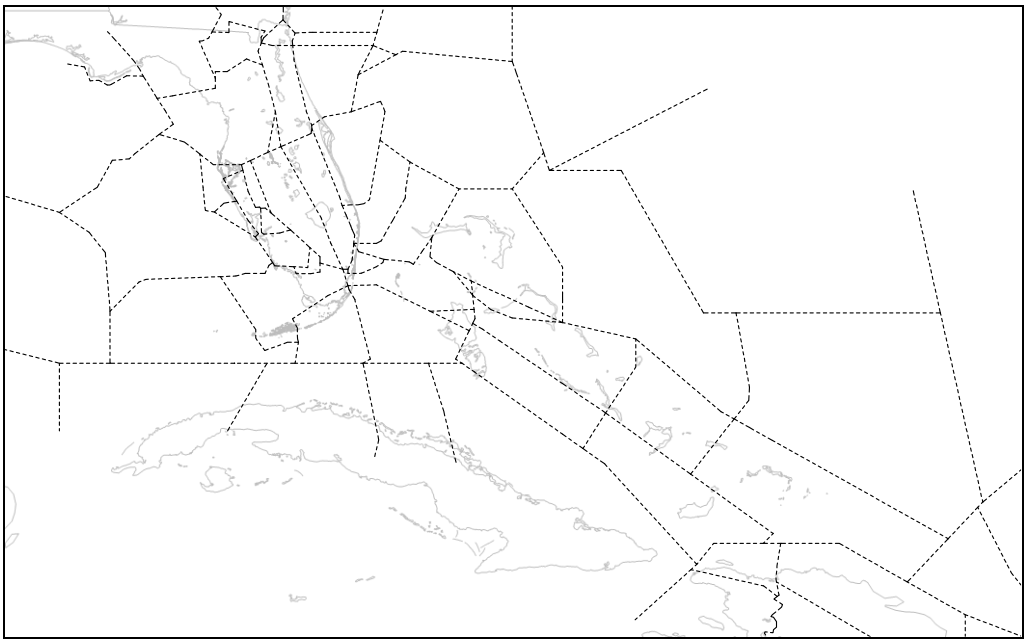

Figure 2 – ZMA/ZMO Geography

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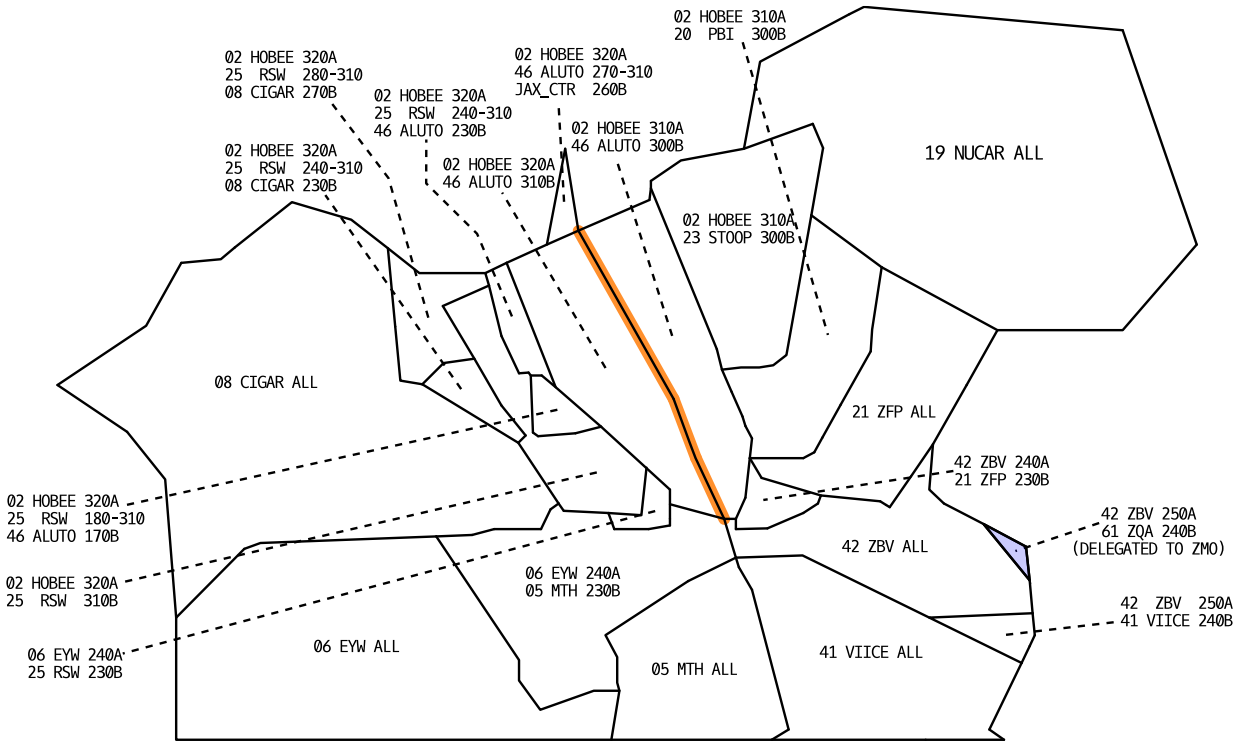

Figure 3 – ZMA Sectorization

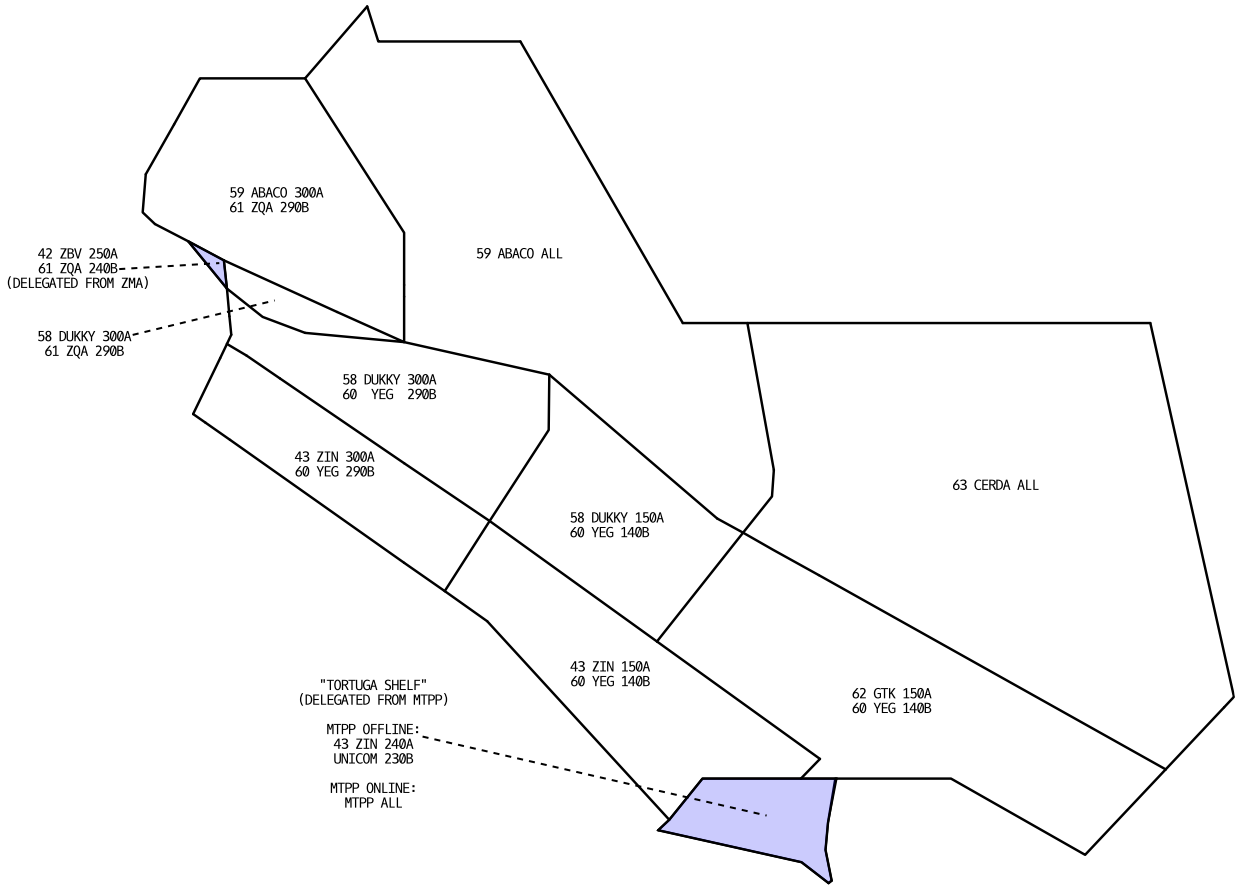

Figure 4 – ZMO Sectorization

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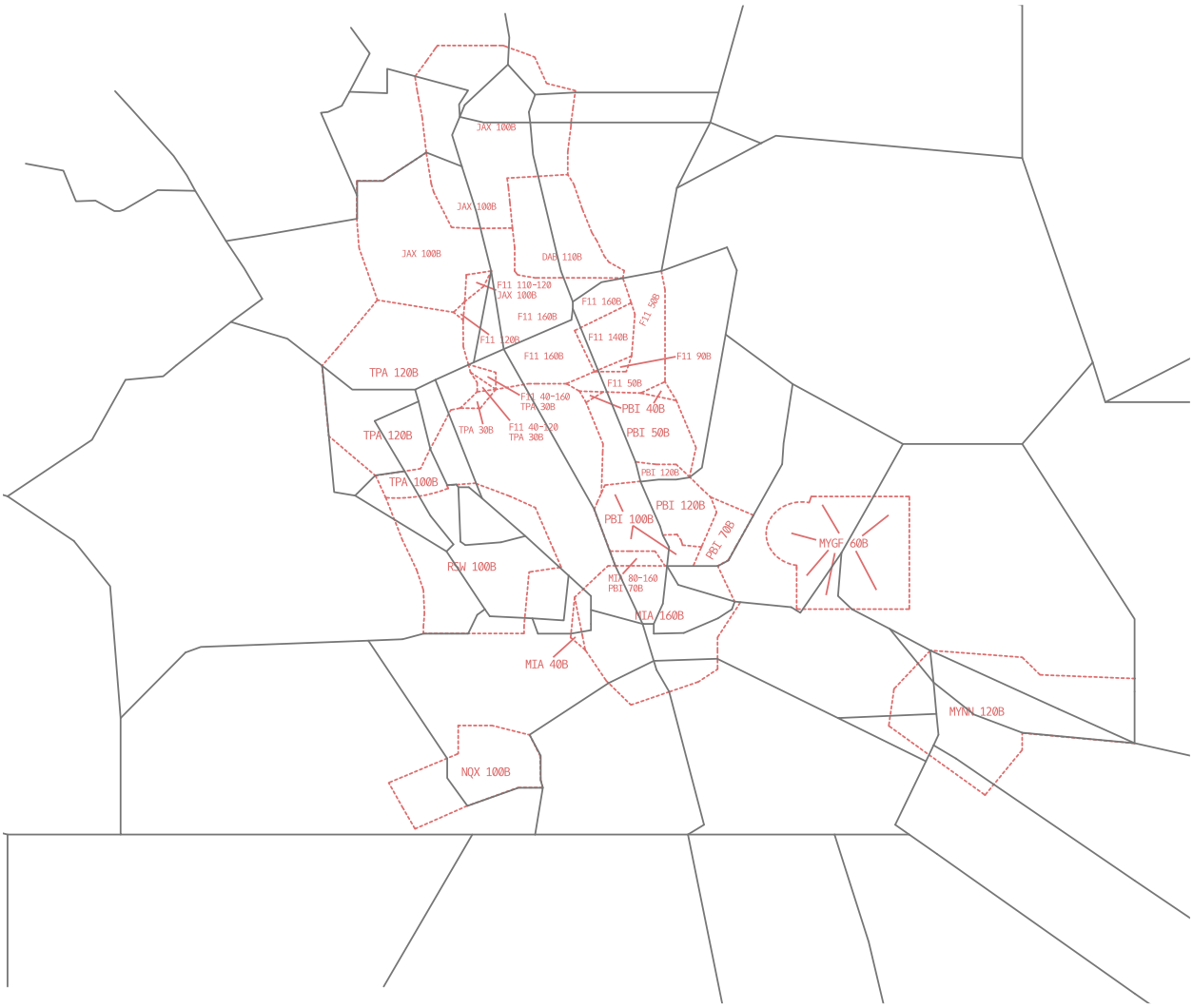

Figure 5 – Approach Controls

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4. General Procedures

The following apply to all en route sectors. Controllers shall be familiar with and comply with them when operating any center position.

a. Within sectors 46, 23, 20, and 02, aircraft east of the Florida centerline shall be assigned ODD altitudes northbound and EVEN altitudes southbound. Outside of this area, normal rules of altitude assignment for direction of flight apply. See Figure 6 to see the area of nonstandard altitude usage. Note the red circled conversions that Sector 21 will perform to deliver aircraft to the next sector at the appropriate altitude.

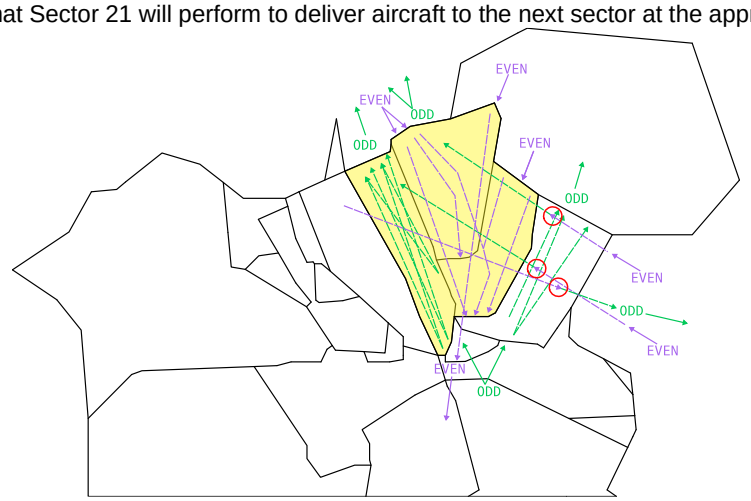

Figure 6 – Special Altitude Assignment Area

b. Adjacent center sectors may use scratchpad (fourth line) data to pass information to the receiving controller in lieu of verbal coordination, provided the information is recorded as indicated by Table *1* below.[1] Note that all clients other than vERAM are limited to either three or four scratchpad characters.

c. "Hard" altitude changes (QZ messages, which amend the cruise altitude) shall be used for all descents into destination airports.

d. When only a single sector of Miami Center is in use, the default sector shall be Sector 46 (ALUTO) for domestic, or Sector 62 (GTK) for oceanic.

   (1) When practical, sign on with low-altitude sector numbers (such as 05) before high-altitude sector numbers (such as 06) when they are combined and being worked by the same controller.

   (2) Open Sector 02 when (and only when) you want arrivals to be forced beneath that sector and into Sectors 20/23/25. Opening Sector 02 will cause ZJX and various ZMA sectors to start forcing arrivals down early in order to support the high-volume center configuration.

e. Miami TRACON may shortcut ("cleared direct") departures no further than the aircraft's filed departure gate without coordination.

f. Navy Key West TRACON may shortcut ("cleared direct") departures no further than CARNU, CANOA, MAXIM, or TADPO without coordination.

DL	Deviating left
D15L	Deviating 15 degrees left
DL/PHK[2]	Deviating left, direct PHK when able
23H	Assigned 230 heading (to any client)
H230	Assigned 230 heading (to vERAM controller)
PH	Assigned present heading
28K	Assigned 280kt speed (to any client)
S280	Assigned 280kt speed (to vERAM controller)
M77	Assigned mach .77
M77+	Assigned mach .77 or greater

Table 1 *- Fourth Line Data Entries*

[1] Note that vERAM's native fourthline headings and speeds will only pass at handoff if both controllers are using CRC. If either controller is using a different client, only the free-form text scratchpad will pass..
[2] Non-CRC and non-vERAM clients will receive only the first 3 or 4 characters of a scratchpad entry.

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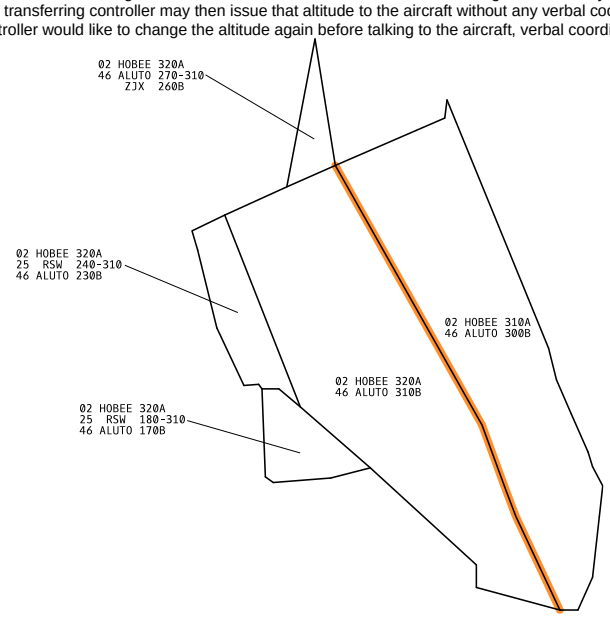
5. EVERGLADES AREA

Sector 46 ALUTO – ZMA Primary Sector

ALUTO is a low-altitude sector, generally at and below FL310 on the west, and FL300 on the east, with a predominantly northbound flow with considerable crossing climbing/descending traffic. ALUTO works departures out of the Miami metro area (routed via BNGOS/BNICE, HURCN/HROCK, FLMGO/FRSBE, ALTNN/AGERS, LIFRR and DORAL), departures out of Palm Beach (via TBIRD, OLAKE and WELLY), departures out of Fort Myers (via CSHEL), departures out of Tampa (via CROWD), and departures out of Orlando (via FSHUN). ALUTO works arrivals to Palm Beach (via VUUDU/TTYLR), Fort Myers (via SHFTY/LBV), Tampa (via BLFRG/BRDGE), and Orlando (via RIDES/GOOFY and PRICY/JOKRS/MINEE). Many different crossing flows are found in this sector, and since most aircraft will be climbing/descending, there is a high potential for conflict.

AIT: When transferring aircraft between HOBEE and ALUTO, the receiving controller may enter an altitude assignment in the data block immediately after accepting the handoff. The transferring controller may then issue that altitude to the aircraft without any verbal coordination before issuing the frequency change. In the event that the receiving controller would like to change the altitude again before talking to the aircraft, verbal coordination with the transferring controller is required.



Standard Callsign	MIA_46_CTR
Alternate Callsign	MIA_AL_CTR
Frequency	135.175
Altimeters	KMIA, KPBI, KFPR, KMCO, KTPA, KRSW, KSEF

Table 2 - ALUTO Sector Details

Note: When Sector 02 (HOBEE) is "inactive", the sector owning the airspace directly beneath it assumes control of HOBEE's altitudes.

SECTOR	DEST	ROUTING	ALTITUDE	SPEED
TPAz	KTPA	**BLFRG#**	DSIDE @ 11,000	TPA N: 250K
		BRDGE#	BRDGE @ 11,000	TPA N: 250K
		BRDGE PIE	BRDGE AOB 11,000	≤ 250K
PBIz	KPBI, KBCT	**VUUDU#**	DES VIA	-----
RSWz	KRSW	**SHFTY#**	MOEMO @ 10,000	RSW W: 250K
MCOz	KMCO	**RIDES#**	DES VIA	-----
		GOOFY#	MCO S: BAIRN @ 11,000 MCO N: BAIRN @ 8,000	-----
		PRICY#	DES VIA	-----
		JOKRS# / MINEE#	[MOANS/ANDRO] @ [J: 13,000 / T: 10,000 / P: 5,000]	MCO N: 250K
	KDAB	JENSN V152 OMN	MCOz BDRY AOB 15,000	-----
23 STOOP	KDAB	MLB V3 OMN	ZMA23 (STOOP) AOB 15,000	-----
ZJX	KSGJ	(any route over F11)	ZJX BDRY AOB FL210	-----
	KGNV	(any route over F11)	ZJX BDRY AOB FL240	-----

Table 3 - ALUTO Routes & Restrictions

FROM	TO	ROUTING	CONTROL GIVEN
25 RSW	46 ALUTO	FERAL **BLFRG#** RSW BRDGE#	30° TURNS
08 CIGAR	46 ALUTO	ALL MCOz LDG TFC	FULL CONTROL
46 ALUTO	TPAz	ALL KSRQ LDG TFC	FULL CONTROL

Table 4 - ALUTO Nonstandard Transfers of Control

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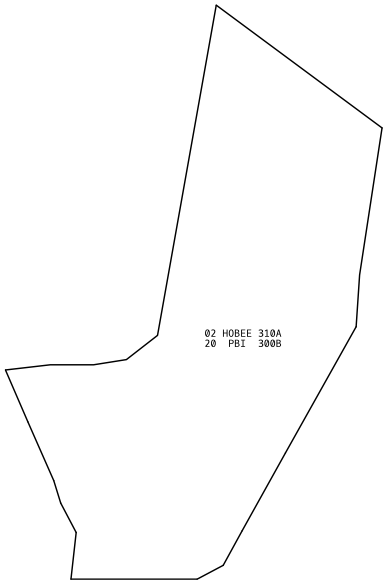
6. COASTAL AREA

Sector 20 PBI (Palm Beach)

PBI is a low-altitude sector at and below FL300. PBI finalizes in-trail spacing for feeds to the Miami TRACON (via CSTAL/ANNEY, BNFSH, and CUUDA) and Palm Beach TRACON (via CPTAN and CLMNT). PBI also works Palm Beach departures (via SLIDZ and MIXAE). This sector is heavily dependent upon receiving a good feed from Sector 23 (STOOP) for Miami area arrivals routed down the coast, since it is such a short distance from where they enter this sector to where they exit. Note that CSTAL and BNFSH traffic need only have the specified in-trail spacing be provided *per route*. Workload permitting, Sector 20 *should* sequence aircraft to stagger the two flows they feed to approach, *but this is not inherently required*.

Apply the special altitude assignment rules laid out in Section 4.a on page 6 (see Figure 6 for a visual depiction).

Deliver aircraft to Sector 21 at EVEN altitudes, and accept aircraft from Sector 21 at ODD altitudes.



Standard Callsign	MIA_20_CTR
Alternate Callsign	MIA_PB_CTR
Frequency	132.15
Altimeters	KPBI, KFPR, MYGF

Table 5 - PBI Sector Details

Note: When Sector 02 (HOBEE) is "inactive", the sector owning the airspace directly beneath it assumes control of HOBEE's altitudes.

SECTOR	DEST	ROUTING	ALTITUDE	SPEED
PBIz	KPBI, KBCT	**CLMNT#**	DES VIA	-----
MIAz	KMIA	**CSTAL#**	DES VIA	
		BNFSH#	DES VIA	
		ANNEY#	MIA E: ANNEY @ 13,000 MIA W: ANNEY @ 11,000	MIA E: 280K MIA W: 250K
	KFLL	**CUUDA#**	DES VIA	
21 ZFP	MYAM, MYAT	ZFP	ZMA21 (ZFP) AOB FL280	-----
	MYGF	-----	DSDG 5,000	-----

Table 6 - PBI Routes & Restrictions

FROM	TO	ROUTING	CONTROL GIVEN
23 STOOP	20 PBI	ALL MIAz LDG TFC	CONTROL TO ISSUE DES VIA

Table 7 - PBI Nonstandard Transfers of Control

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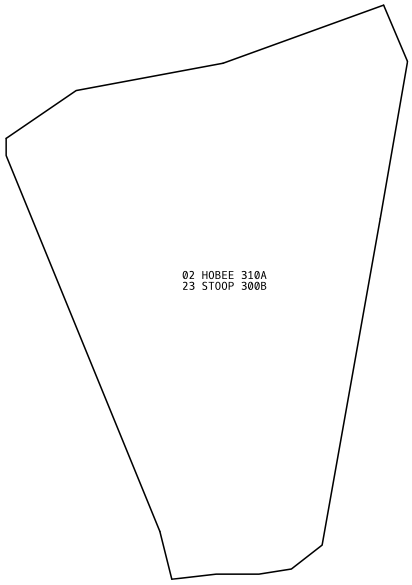
Sector 23 STOOP

STOOP is a low-altitude sector at and below FL300 with a predominantly southbound flow. STOOP provides initial in-trail spacing for feeds to the Miami Area (via CSTAL/ANNEY and CUUDA), and finalizes in-trail spacing to Palm Beach (via CPTAN) and Orlando (via ALYNA). STOOP also works Orlando departures (via MZULO and DDANY). Occasional westbound Fort Myers arrivals (via HIBAC SHFTY#) are a potential issue with blocking the descents of the southbound arrivals. These Fort Myers arrivals usually do not need the (very aggressive) real-world LOOOU @ FL240 restriction, so coordinate with Sector 46 (ALUTO) if traffic conditions suggest they might want them lower than FL300. When Sector 02 (HOBEE) opens above STOOP, ZJX is required to force all “landing” aircraft down beneath HOBEE, to enter STOOP directly. Sector 19 (NUCAR) will do the same.

Apply the special altitude assignment rules laid out in Section 4.a on page 6 (see Figure 6 for a visual depiction).

Deliver aircraft to Sector 20 at EVEN altitudes, and accept aircraft from Sector 20 at ODD altitudes.



Standard Callsign	MIA_23_CTR
Alternate Callsign	MIA_ST_CTR
Frequency	126.95
Altimeters	KFPR, KMCO

Table 8 - STOOP Sector Details

Note: When Sector 02 (HOBEE) is “inactive”, the sector owning the airspace directly beneath it assumes control of HOBEE’s altitudes.

SECTOR	DEST	ROUTING	ALTITUDE	SPEED
PBIz	KMIA, KFLL	**TARPN#**	RONQU @ 12,000	-----
		BLUFI#	DALOO @ 12,000	-----
	KPBI, KBCT	**CPTAN#**	DES VIA	-----
		JESTR#	DES VIA	-----
		STOOP#	STOOP @ 10,000	-----
		MLB#	STOOP @ 10,000	-----
MCOz	KMCO	**ALYNA#**	SURFR @ 14,000	-----
	KDAB	MLB V3 OMN	MLB AOB 15,000	-----
	KTPA	HIBAC **DADES#**	ZINGR @ 16,000	-----
20 PBI	KMIA	**CSTAL#**	XPRSO AOB FL260	-----
		ANNEY#	30 PBI AOB FL260	-----
	KFLL	**CUUDA#**	COLAT AOB FL240	-----
	MYGF	-----	ZMA20 (PBI) AOB 15,000	-----
46 ALUTO	KRSW	HIBAC **SHFTY#**	ZMA46 (ALUTO) AOB FL300	-----
ZJX	KSGJ	(any route over F11)	ZJX BDRY AOB FL210	-----
	KGNV	(any route over F11)	ZJX BDRY AOB FL230	-----

Table 9 - STOOP Routes & Restrictions

FROM	TO	ROUTING	CONTROL GIVEN
23 STOOP	20 PBI	ALL MIAz LDG TFC	CONTROL TO ISSUE DES VIA

Table 10 - STOOP Nonstandard Transfers of Control

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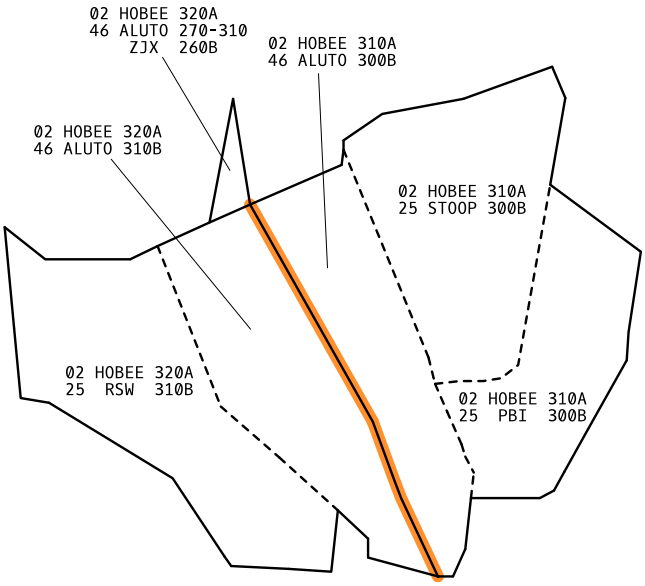
7. ATLANTIC AREA

Sector 02 HOBEE

HOBEE is a high-altitude sector, at and above FL320 on the west and FL310 on the east, overlying Sector 23 (STOOP), Sector 20 (PBI), Sector 46 (ALUTO), and Sector 25 (RSW). HOBEE serves primarily to reduce the workload of those sectors. When open, all arrival traffic is generally forced beneath HOBEE, such that HOBEE will work almost exclusively long-range overflight traffic. Only arrivals to Jacksonville or Bahamian destinations require any descent.

AIT: When transferring aircraft between HOBEE and ALUTO, the receiving controller may enter an altitude assignment in the data block immediately after accepting the handoff. The transferring controller may then issue that altitude to the aircraft without any verbal coordination before issuing the frequency change. In the event that the receiving controller would like to change the altitude again before talking to the aircraft, verbal coordination with the transferring controller is required.



Standard Callsign	MIA_02_CTR
Alternate Callsign	MIA_2_CTR / MIA_HO_CTR
Frequency	119.825
Altimeters	NONE

Table 11 - HOBEE Sector Details

Note: When Sector 02 (HOBEE) is "inactive", its airspace is divided up amongst the lows beneath it.

SECTOR	DEST	ROUTING	ALTITUDE	SPEED
ZJX	JAX+	HAINY **QUBEN#** ORL POGIE#	ZJX BDRY AOB FL310	-----
	KTLH	-----	ZJX BDRY AOB FL340	
20 PBI	MYA*	ZFP	ZMA20 (PBI) AOB FL310	-----
21 ZFP	MYNN	ZFP BR63V ZQA	ZMA21 AOB FL340	-----
46 ALUTO	KSGJ KGNV	(any route over F11)	DSDG TO BOTTOM OF AIRSPACE	-----

Table 12 - HOBEE Routes & Restrictions

FROM	TO	ROUTING	CONTROL GIVEN
----- NONE -----			

Table 13 - HOBEE Nonstandard Transfers of Control

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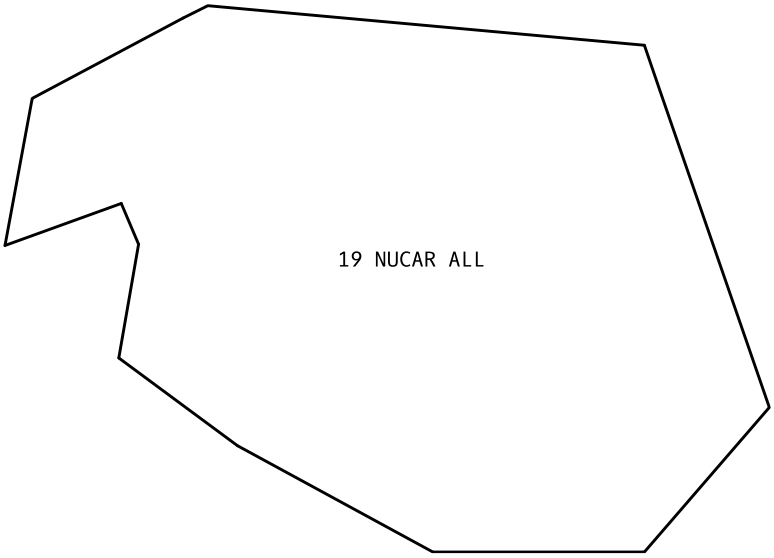
Sector 19 NUCAR

NUCAR is a non-stratified sector owning all altitudes, located in the very northeast corner of ZMA airspace. NUCAR works aircraft transiting between the Atlantic Routes (ARs), the Bahamas, the Ocean area, central to southern Florida, and deep-water transatlantic routes (via SUMRS/SNAGY). Aircraft generally remain in level flight in this sector, and while volume can become high, complexity usually remains very low.

For aircraft entering New York Oceanic airspace, NUCAR must provide same-altitude aircraft to ZNY in trail with appropriate nonradar separation already established by the airspace boundary. This would be 10 minutes or 20 miles, after assigning speeds which ensure the trailing aircraft is at the same or slower speed.



Standard Callsign	MIA_19_CTR
Alternate Callsign	MIA_NU_CTR
Frequency	123.675
Altimeters	-----

Table 14 - NUCAR Sector Details

SECTOR	DEST	ROUTING	ALTITUDE	SPEED
20 PBI	KPBI / KBCT	DULEE **CLMNT#**	CRANS AOB FL260	-----
	KFLL	MAJIK **CUUDA#**	AYBID AOB FL300	-----
	KMIA	HOAGG **BNFSH#**	JORAY AOB FL300	-----
23 STOOP	KMCO	HIBAC **ALYNA#**	HIBAC AOB FL280	-----
	KRSW	HIBAC **SHFTY#**	*ONLY IF ZMA02 (HOBEE) OPEN: HIBAC AOB FL300*	-----
59 ABACO	TJSJ	NUCAR Y355 PLING SAALR	-----	-----
61 ZQA	MYNN	AR3 ZQA DCT	BDRY AOB FL280	-----

Table 15 - NUCAR Routes & Restrictions

FROM	TO	ROUTING	CONTROL GIVEN
----- NONE -----			

Table 16 - NUCAR Nonstandard Transfers of Control

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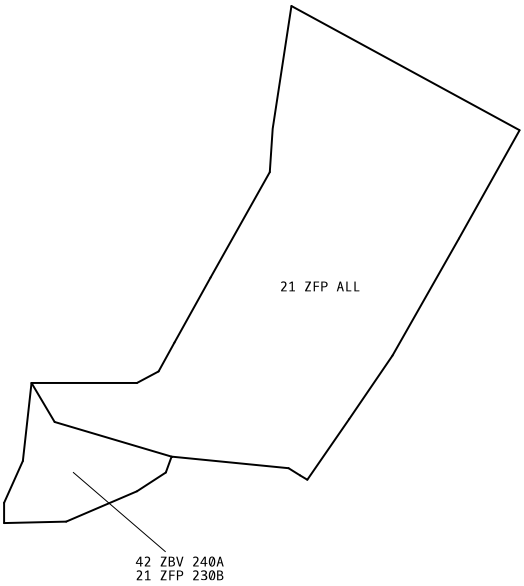
Sector 21 ZFP (Freeport)

ZFP is a small multipurpose sector working various crossing flows for a short amount of time. ZFP serves to work northbound departures from the Miami area toward the ARs (via FOLZZ/FEALX), Nassau arrivals to Nassau Center (via ZFP), arrivals to Freeport Approach, high-altitude overflights from the Everglades area to the Ocean area, and overflight or landing traffic transitioning from the Ocean area to the Everglades area. Rare arrivals to the Miami area routed over ZFP may necessitate coordination with Sector 61 ZQA to facilitate some tight-quarters swapping out with the departure flow, but usually other sectors will move aircraft off of these undesirable routes.

Deliver aircraft to Sectors 46, 20 and 02 at ODD altitudes, and accept aircraft from these sectors at EVEN altitudes. These sectors use the special altitude assignment rules laid out in Section 4.a on page 6 (see Figure 6 for a visual depiction of altitude usage).

When workload permits and alternative routings are feasible, do not allow aircraft to remain on northbound routes through Sectors 20/23. Aircraft should be routed up the ARs to Sector 19, or over PBI and into Sector 46.



Standard Callsign	MIA_21_CTR
Alternate Callsign	MIA_ZF_CTR
Frequency	133.4
Altimeters	KMIA, KPBI, MYGF

Table 17 - ZFP Sector Details

SECTOR	DEST	ROUTING	ALTITUDE	SPEED
PBIz	KPBI	MAHHI **MAHHI#**	MAHHI @ 5,000	---
MYGFz	MYGF	-----	MYGF APP BDRY @ 5,000	-----
20 PBI	KMIA	[FIROL / REPEE / MUNRO] **BNFSH#**	BICAP @ FL190	-----
		SNSBK ANNEY#	SNSBK AOB 16,000	-----
	KFLL	[FIROL / REPEE / MUNRO] **CUUDA#**	PLAID @ 8,000	-----
MYGFz	MYGF	-----	MYGF APP BDRY @ 5,000	-----
46 ALUTO	KMCO	LAIZY **RIDES#**	LAIZY AOB FL290	-----
42 ZBV	KFLL	JEKER **OLAHS#**	ZMA42 (ZBV) AOB FL180 DSDG 14,000	-----
61 ZQA	MYNN	ZFP BR63V ZQA	ZFP AOB FL290	-----
	MYAM / MYAT	ZFP	ZFP AOB FL190	-----

Table 18 - ZFP Routes & Restrictions

FROM	TO	ROUTING	CONTROL GIVEN
----- NONE -----			

Table 19 - ZFP Nonstandard Transfers of Control

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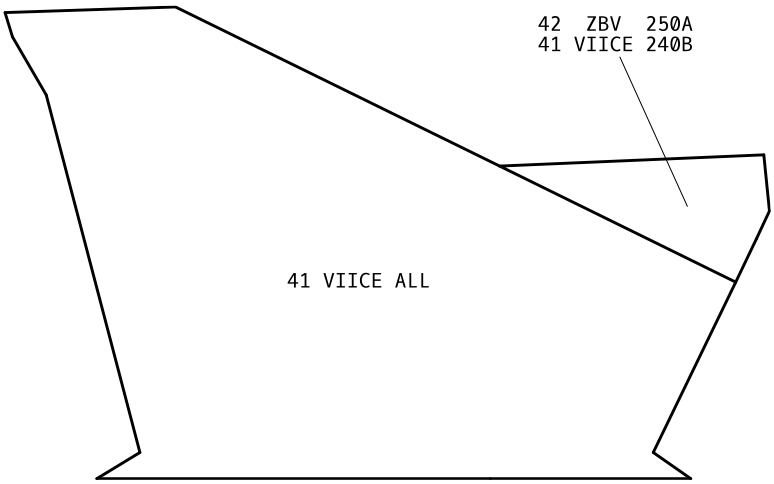
8. Caribbean Area

Sector 41 VIICE

VIICE is a relatively small sector bordering Miami Approach, Nassau Approach, and Havana Center. VIICE is frequently combined with Sector 42, Sector 05, or both. VIICE provides service to Miami arrivals (via VIICE/FOWEE), Nassau arrivals (via SANNS), Miami departures (via GWAVA), and Nassau Departures (via NICKO). The majority of transiting Caribbean traffic is routed through Sector 42, so this sector is typically pretty straightforward, though it can experience high traffic volume.



Standard Callsign	MIA_41_CTR
Alternate Callsign	MIA_VI_CTR
Frequency	135.6
Altimeters	KMIA, MYNN

Table 20 - VIICE Sector Details

Sector	Dest	Routing	Altitude	Speed
MIAz	KMIA	**VIICE#**	DES VIA	-----
		FOWEE#	MIA E: JUNUR @ 12,000 MIA W: JUNUR @ 10,000	MIA E: ≤ 280K MIA W: ≤ 250K
MYNNz	MYNN	SANNS ZQA	SANNS @ 11,000	-----
05 MTH	KEYW	JANKA	ZMA05 (MTH) AOB FL280	-----
42 ZBV	KFLL	BORDO **OLAHS#**	WEBAB AOB FL250	-----
		ZBV DEKAL#	15 ZBV AOB 12,000	-----
MUFH	MUHA	MUHA E: TANIA **JUGAR#** MUHA W: TANIA **AFFIL#**	-----	-----

Table 21 - VIICE Routes & Restrictions

From	To	Routing	Control Given
41 VIICE	MIAz	JUNUR ATA[3]	DESCENT & 40° DEGREE TURNS

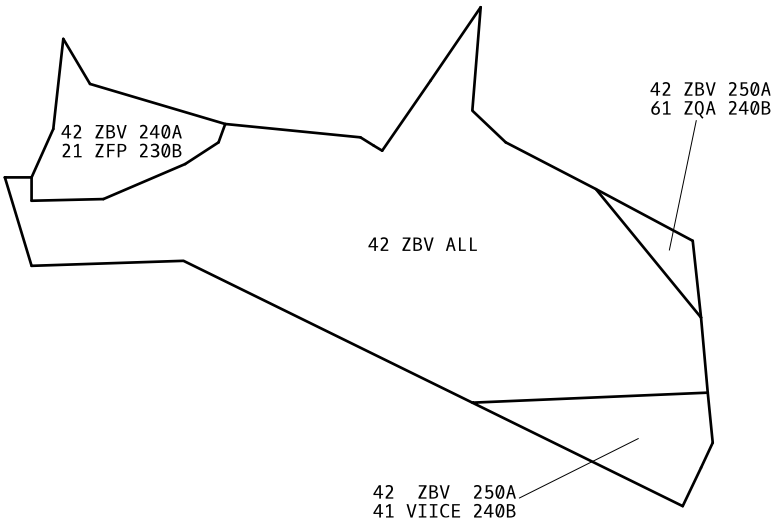
Table 22 - VIICE Nonstandard Transfers of Control

[3] The JUNUR ATA includes VIICE arrivals to KMIA.

For VATSIM simulation purposes only. No affiliation with FAA.

Sector 42 ZBV (Bimini)

ZBV is very narrow sector handling the Fort Lauderdale to Nassau corridor. ZBV works Fort Lauderdale arrivals (via OLAHS/KYAKS), Nassau arrivals (via RAJAY), Nassau Departures (via MAJUR), Miami departures (via KLADA/HUSIL), Fort Lauderdale departures (via SNAPR/REGAE), and high-altitude overflights transitioning between the Ocean area and Everglades area.



Standard Callsign	MIA_42_CTR
Alternate Callsign	MIA_ZB_CTR
Frequency	133.85
Altimeters	-----

Table 23 - ZBV Sector Details

SECTOR	DEST	ROUTING	ALTITUDE	SPEED
MIAz	KFLL	**OLAHS# (Jets)**	DES VIA	-----
		KYAKS# (Props)	DES VIA	
		ZBV DEKAL#	DEKAL @ [J: 6,000 / P: 4,000]	
MYNNz	MYNN	RAJAY ZQA	RAJAY @ 11,000	250K[4]
21 ZFP	KPBI	MAHHI **MAHHI#**	ZMA21 BDRY AOB FL180	-----
46 ALUTO	KTPA	OCTAL **BLFRG#**	OCTAL AOB FL290	-----
	KMCO	OCTAL **RIDES#**	OCTAL AOB FL290	-----
	KRSW	OCTAL PHK LBV RSW	OCTAL AOB FL250	-----

Table 24 - ZBV Routes & Restrictions

FROM	TO	ROUTING	CONTROL GIVEN
----- NONE -----			

Table 25 - ZBV Nonstandard Transfers of Control

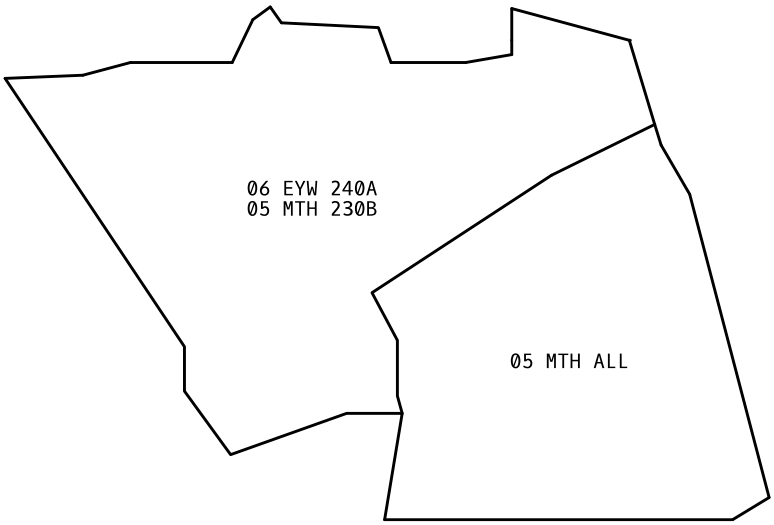
[4] Speed may be omitted if 15MIT or greater is already being provided.

For VATSIM simulation purposes only. No affiliation with FAA.

9. WESTSIDE COASTAL AREA

Sector 05 MTH (Marathon)

MTH is a small sector covering the Florida Keys. MTH finalizes in-trail spacing on Miami and Fort Lauderdale arrivals (via SNDBR and BHHIA), Key West arrivals (via GIGIH and JANKA) and Fort Myers arrivals (via KARTR/GOODY), as well as Miami area departures (via GLADZ and MAYNR), northbound/eastbound departures out of Key West, and departures out of Fort Myers (via MARCI). Approach and departure control services are provided by this sector into and out of KMTH (non-towered).



Standard Callsign	MIA_05_CTR
Alternate Callsign	MIA_MT_CTR
Frequency	132.7
Altimeters	KMIA, KMTH, KEYW, KRSW

Table 26 - MTH Sector Details

SECTOR	DEST	ROUTING	ALTITUDE	SPEED
MIAZ	KMIA / KFLL	**SNDBR#**	DES VIA	-----
		DVALL#	MIA E: WEVER @ 10,000 MIA W: WEVER @ 14,000	250K
RSWz	KRSW	V539 RSW	RSWz BDRY AOB 10,000	-----
		GOODY DCT		
NQXz	KEYW	GIGIH	GIGIH @ 10,000	-----
		JANKA	JANKA @ 10,000	-----

Table 27 - MTH Routes & Restrictions

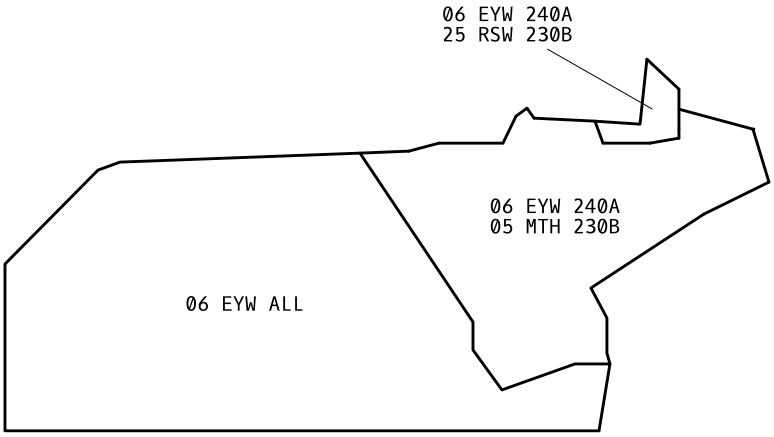
FROM	TO	ROUTING	CONTROL GIVEN
05 MTH	MIAz	WEVER ATA	DESCENT & 40° DEGREE TURNS
06 EYW	05 MTH	**SNDBR#**	CONTROL TO ISSUE DES VIA

Table 28 - MTH Nonstandard Transfers of Control

For VATSIM simulation purposes only. No affiliation with FAA.

Sector 06 EYW (Key West)

EYW is a high-altitude sector in the southwestern corner of Miami Center airspace. EYW provides initial spacing for arrivals to Fort Lauderdale and Miami (via SNDBR/BHHIA), and to Havana (via CANOA or MAXIM). Due to the typically low volume and low complexity of this sector, it is frequently combined with Sector 08 (CIGAR), in order to alleviate the burden of giving such a large area of responsibility to controllers working the major Miami arrival flows in Sector 05 (MTH) and Sector 25 (RSW).



Standard Callsign	MIA_06_CTR
Alternate Callsign	MIA_EY_CTR
Frequency	132.2
Altimeters	KEYW

Table 29 - EYW Sector Details

SECTOR	DEST	ROUTING	ALTITUDE	SPEED
05 MTH	KMIA / KFLL	**SNDBR#**	CONCK AOB FL250	-----
		EYW DVALL#	DSDG TO FL240	-----
	KRSW	V539 RSW	ABEAM EYW AOB FL240	-----
		GOODY DCT		
25 RSW	KTPA	FERAL **BLFRG#** RSW BRDGE#	ZMA25 (RSW) AOB FL300[5]	-----
	KMCO	FERAL **JOKRS#** RSW MINEE#		
MUFH	MUHA	CANOA **ENTRY#**[6]	CANOA AOB FL260	-----

Table 30 - EYW Routes & Restrictions

FROM	TO	ROUTING	CONTROL GIVEN
06 EYW	05 MTH	**SNDBR#**	CONTROL TO ISSUE DES VIA

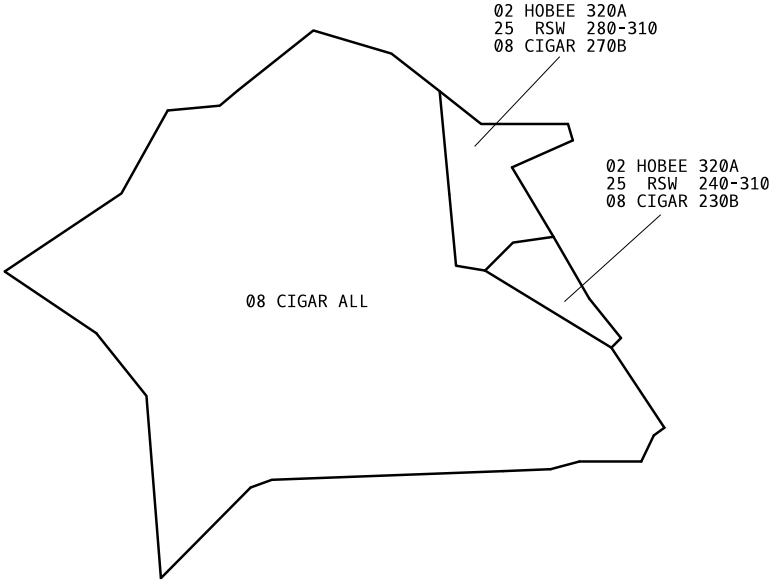
Table 31 - EYW Nonstandard Transfers of Control

[5] TPA and MCO traffic may be stacked to Sector 25 only if the TPA is on the bottom.
[6] In the rare event that Havana is landing west, assign “MAXIM JUGAR#” instead. See MUFH LOA.

For VATSIM simulation purposes only. No affiliation with FAA.

Sector 08 CIGAR

CIGAR is a large sector bordering Jacksonville Center, Houston Center, Tampa Approach, and Ft Myers Approach. A small, low-altitude shelf exists with CIGAR and Sector 25 (RSW) in order to alleviate RSW of the need to talk to any Tampa, Orlando, and Ft Myers traffic. CIGAR provides initial in-trail spacing for arrivals to Miami and Fort Lauderdale (via GAWKS) and Orlando (via OSPRY), as well as finalizing in-trail spacing to Tampa (via RAYZZ). CIGAR also works westbound Tampa departures (via KNOST/DORMR). Note that CIGAR does not work MCO traffic all the way in—they will be handed off to Sector 46 (ALUTO) for sequencing with the JOKRS arrivals.



Standard Callsign	MIA_08_CTR
Alternate Callsign	MIA_CI_CTR
Frequency	133.9
Altimeters	-----

Table 32 - CIGAR Sector Details

Note: When Sector 02 (HOBEE) is "inactive", the sector owning the airspace directly beneath it assumes control of HOBEE's altitudes.

SECTOR	DEST	ROUTING	ALTITUDE	SPEED
TPAz	KTPA	**RAYZZ#**	DES VIA	-----
25 RSW	KMIA	GAWKS **FROGZ#**	GAWKS AOB FL310[7]	-----
	KFLL	GAWKS **TEEKY#**		-----
	KPBI / KBCT	GAWKS **VUUDU#**	GAWKS AOB FL290	-----
	KAPF	[PLYER / REMIS] **PIKKR#**	ZMA25 (RSW) AOB FL270	-----
	KRSW	[BAGGS / CODGR] **TYNEE#**	ZMA25 (RSW) AOB FL230	-----
46 ALUTO	KMCO	**PRICY#**	DES VIA[8]	-----
		MINEE#	GUMMY AOB 17,000	
ZJX	KDAB	WHITL KNEED V152 OMN	WHITL AOB FL310	-----

Table 33 - CIGAR Routes & Restrictions

FROM	TO	ROUTING	CONTROL GIVEN
08 CIGAR	46 ALUTO	ALL MCOz LDG TFC	FULL CONTROL

Table 34 - CIGAR Nonstandard Transfers of Control

[7] MIA/FLL traffic may be stacked to Sector 25 only if the FLL is on the bottom.
[8] Sector 08 issues D/V, but shall **transfer communications to Sector 46 prior to DECKS** (abeam PIE).

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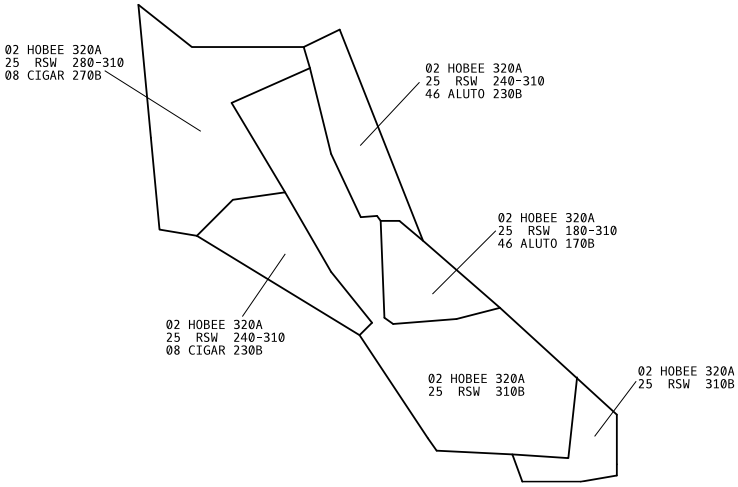
Sector 25 RSW (Fort Myers)

RSW is a fairly complex low-altitude sector with numerous parallel southbound descending flows. RSW works arrivals to Fort Myers (via TYNEE/JOSFF), Naples (via PIKKR/ZEILR), the Miami area (via FROGZ/PALMZ and TEEKY/FORTL), and Palm Beach (via VUUDU). RSW also works Tampa departures routed via GANDY, and a few overflights, but the meat of this sector is in the sequencing of the FROGZ and TEEKY arrivals into Miami and Fort Lauderdale. When Sector 02 (HOBEE) opens above RSW, ZJX is required to force all "landing" aircraft down beneath HOBEE, to enter RSW directly. Sector 08 (CIGAR) will do the same.



Standard Callsign	MIA_25_CTR
Alternate Callsign	MIA_RS_CTR
Frequency	128.225
Altimeters	KMIA, KRSW, KTPA

Table 35 - RSW Sector Details

Note: When Sector 02 (HOBEE) is "inactive", the sector owning the airspace directly beneath it assumes control of HOBEE's altitudes.

SECTOR	DEST	ROUTING	ALTITUDE	SPEED
RSWz	KRSW	**TYNEE#** / JOSFF#	TYNEE @ 11,000	-----
	KAPF	**PIKKR#** / ZEILR#	ZEILR @ 11,000	
MIAz	KMIA	**FROGZ#**	DES VIA	-----
		PALMZ#	COOFS @ 8,000	-----
	KFLL	**TEEKY#**	DES VIA	-----
		FORTL#	BANTY @ 5,000	-----
05 MTH	KEYW	GIGIH	ZMA05 (MTH) AOB FL230	-----
46 ALUTO	KTPA	FERAL **BLFRG#**	15 BLFRG AOB FL180	-----
		RSW BRDGE#	25 ROGAN AOB FL180	
	KMCO	FERAL **JOKRS#**	CSHEL AOB FL230	-----
		RSW MINEE#	35 DOWNN AOB FL230	
	KPBI, KBCT	**VUUDU#**	DES VIA	-----

Table 36 - RSW Routes & Restrictions

FROM	TO	ROUTING	CONTROL GIVEN
25 RSW	46 ALUTO	RSW DEAKK/BRDGE#	30° TURNS
25 RSW	MIAz	WORPP ATA	DESCENT & 40° DEGREE TURNS

Table 37 - RSW Nonstandard Transfers of Control

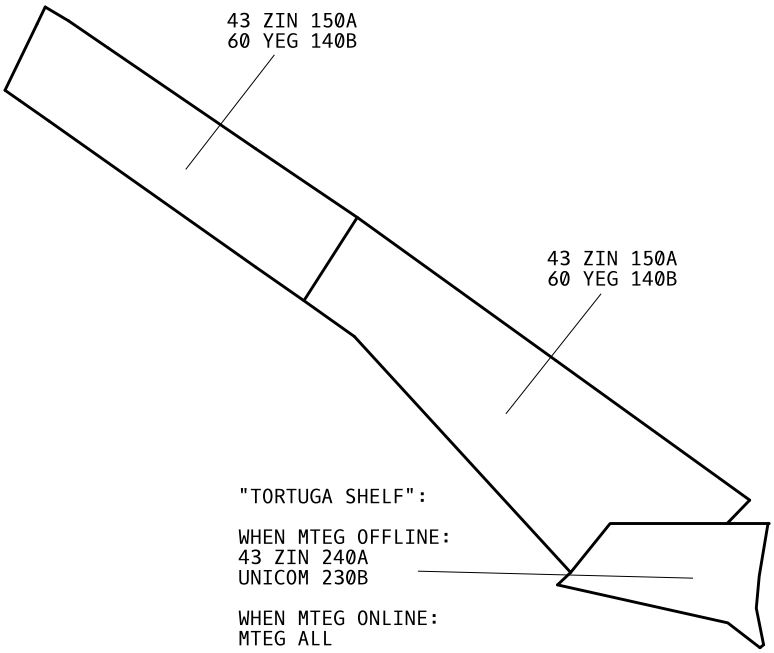
For VATSIM simulation purposes only. No affiliation with FAA.

10. OCEAN AREA

Sector 43 ZIN (Great Inagua)

ZIN is the sector owning the southern corridor of the Ocean area. By design, this sector should contain all (and for the most part, *only*) northwestbound arrivals to KMIA.

The lower altitudes beneath this sector are covered fully by Sector 60, so there should be minimal slow low-altitude aircraft, and zero top-down responsibilities.



Standard Callsign	ZMO_43_CTR
Alternate Callsign	ZMO_ZI_CTR
Frequency	123.775
Altimeters	-----

Table 38 - ZIN Sector Details

SECTOR	DEST	ROUTING	ALTITUDE	SPEED
41 VIICE	KMIA	**MADIZ VIICE#** MADIZ FOWEE FOWEE#	-----	-----
MTEG	MTPP	-----	MTEG ONLINE: BDRY AOB FL270 MTEG OFFLINE: LEAVE TORTUGA AOB FL240	-----
	MTCH	-----	MTEG BDRY AOB FL150[9]	-----

Table 39 - ZIN Routes & Restrictions

FROM	TO	ROUTING	CONTROL GIVEN
----- NONE -----			

Table 40 - ZIN Nonstandard Transfers of Control

[9] Note that this is genuinely “Flight Level 1-5-0”, and not 15,000 feet, due to lack of nearby altimeters.

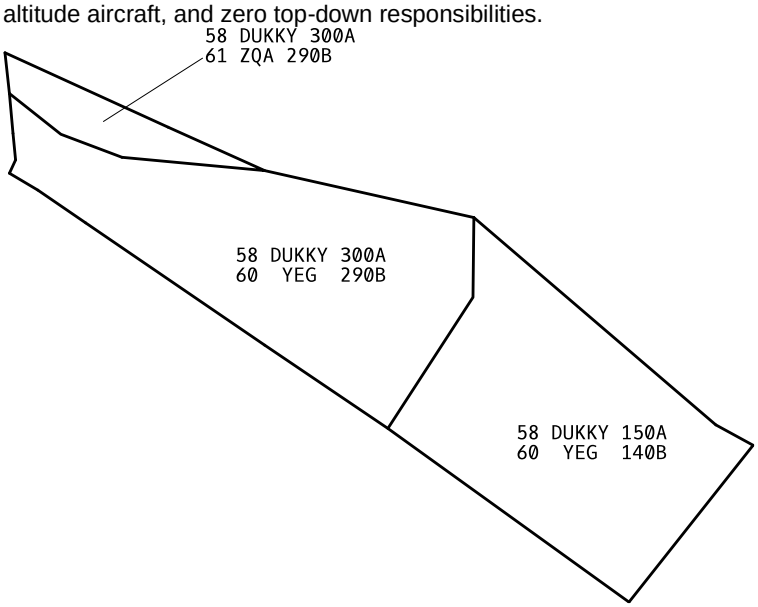
For VATSIM simulation purposes only. No affiliation with FAA.

Sector 58 DUKKY

DUKKY is the sector extending from Nassau, Bahamas to the southeast. Traffic flows in both directions in this sector, with the most common aircraft being Fort Lauderdale arrivals (via OLAHS#), southeastbound departures (such as KLADA/HUSIL, SNAPR/GABOW), and transiting overflight traffic. Nassau arrival traffic will also be seen occasionally over SEAAN.

For planning and sequencing purposes, Sector 58 must send an automated point out to Sector 43 for any FLIPR arrivals, no later than when the aircraft enters Sector 58 airspace. Normally, these aircraft should be routed over BOOZY from Sector 62 directly into Sector 43, but when this doesn't happen, Sector 58 must do this in order to make Sector 43 aware of the aircraft's presence.

The lower altitudes beneath this sector are covered fully by Sectors 60 and 61, so there should be minimal slow low-altitude aircraft, and zero top-down responsibilities.



Standard Callsign	ZMO_58_CTR
Alternate Callsign	ZMO_DU_CTR
Frequency	124.525
Altimeters	-----

Table 41 - DUKKY Sector Details

SECTOR	DEST	ROUTING	ALTITUDE	SPEED
42 ZBV	KFLL (J)	SEAAN PEACH **OLAHS#**	PEACH AOB FL300	-----
	KPBI	SEAAN PEACH **MAHHI#**	PEACH AOB FL300	-----
	KTPA	SUMAC Y280 OCTAL **BLFRG#**	-----	-----
	KMCO	SUMAC Y280 OCTAL **RIDES#**	-----	-----
	KRSW	SUMAC Y280 OCTAL PHK LBV	-----	-----
62 GTK	TJSJ	**HAGIT Y290 BEANO** GTK A555 IDAHO RTE6 SJU	-----	-----
	TNCM	**HAGIT Y290 SLUGO** GTK A555 IDAHO RTE6 SJU SLUGO	-----	-----
60 YEG	MYNN	SEAAN ZQA	SHELF BDRY AOB FL280	-----
61 ZQA	MYG*, MYA*	-----	ZMO61 BDRY AOB FL300	-----
	KFLL (P)	SEAAN PEACH **KYAKS#**	SEAAN AOB FL300	-----

Table 42 - DUKKY Routes & Restrictions

FROM	TO	ROUTING	CONTROL GIVEN
58 DUKKY	60 YEG	ALL LDG TFC	TURNS & DESCENT

Table 43 - DUKKY Nonstandard Transfers of Control

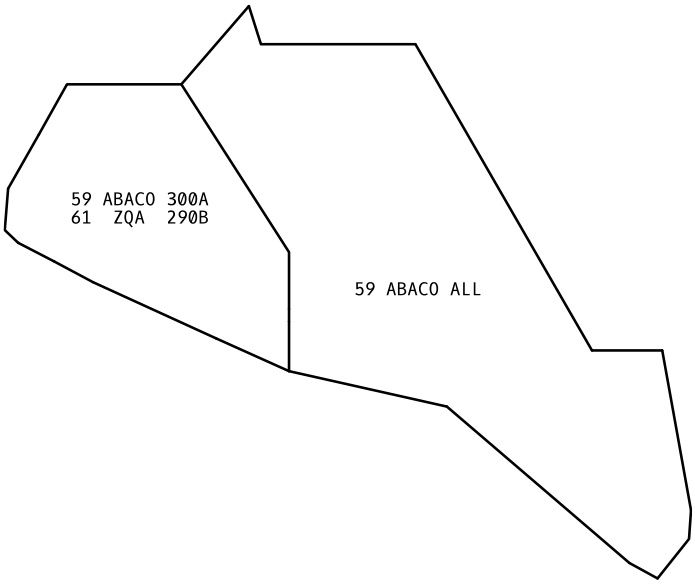
For VATSIM simulation purposes only. No affiliation with FAA.

Sector 59 ABACO

ABACO is the most northern sector of the Ocean area, over the northern Bahamas. This sector works mostly overflight traffic, with some aircraft descending into Sectors 60 and 61 (via DAAST and BAAGR).

When workload permits and alternative routings are feasible, do not allow aircraft to remain routed over ZFP who land within ZMA, F11, or KJAX. These aircraft shall be rerouted through Sector 58 over ZQA.

Sector 21 (ZFP) will do all conversion of altitudes to comply with the Special Altitude Assignment Area described in Section 4.a on page 6. Sector 59 (ABACO) shall deliver aircraft to Sector 21 using *normal altitude for direction of flight rules* (all westbound aircraft at EVEN altitudes).



Standard Callsign	ZMO_59_CTR
Alternate Callsign	ZMO_AB_CTR
Frequency	134.8
Altimeters	-----

Table 44 – ABACO Sector Details

SECTOR	DEST	ROUTING	ALTITUDE	SPEED
21 ZFP	KMIA	[FIROL / REPEE / MUNRO] **BNFSH#** ZFP SNSBK ANNEY#	REPEE AOB FL240	-----
	KFLL	[FIROL / REPEE / MUNRO] **CUUDA#**	REPEE AOB FL200[10]	-----
	KPBI	ZFP BR21V PBI	ZFP AOB FL200[10]	-----
61 ZQA	MYNN	BAAGR BARTS AR3 ZQA	BAAGR AOB FL280	-----
		DAAST M329 ZQA	DAAST AOB FL240	-----
	MYGF	BAAGR ZFP	BAAGR AOB FL280	-----
		DAAST ZFP	DAAST AOB FL280	-----
	MYAM, MYAT	-----	BDRY AOB FL180	-----
63 CERDA	TJSJ	**RENAH Y355 PLING SAALR**	-----	-----
	TNCM	**RENAH Y355 SLUGO**	-----	-----

Table 45 - ABACO Routes & Restrictions

FROM	TO	ROUTING	CONTROL GIVEN
----- NONE -----			

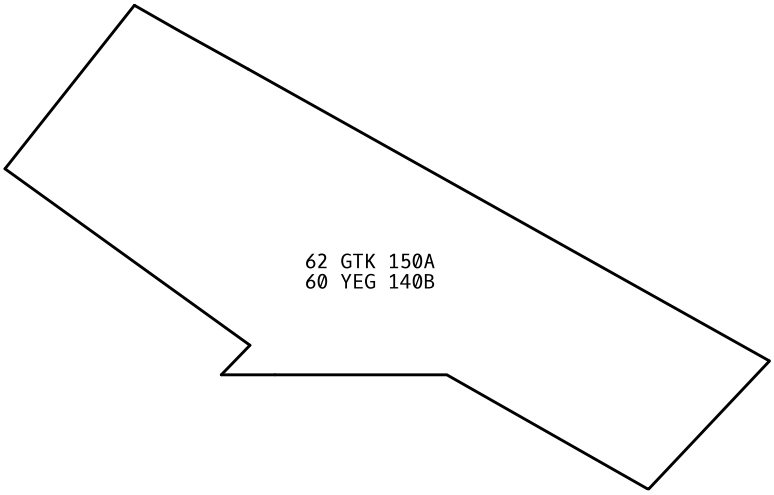
Table 46 - ABACO Nonstandard Transfers of Control

[10] KFLL and KPBI traffic may be delivered to Sector 21 stacked at FL180 and FL200, with the KFLL on the bottom.

For VATSIM simulation purposes only. No affiliation with FAA.

Sector 62 GTK (Grand Turk) – ZMO Primary Sector

GTK is the most southern sector of the Ocean area, over Turks and Caicos. GTK adjoins Port Au Prince FIR and Santo Domingo FIR to the south, and San Juan to the east. This sector primarily sees overflight traffic transiting between Florida, deep-water transatlantic routes, San Juan, Juliana, St Thomas, Nassau, Santo Domingo and Punta Cana.



Standard Callsign	ZMO_62_CTR
Alternate Callsign	ZMO_GT_CTR
Frequency	132.3
Altimeters	-----

Table 47 - GTK Sector Details

SECTOR	DEST	ROUTING	ALTITUDE	SPEED
43 ZIN	KMIA	BOOZY Y261 MADIZ [**VIICE#** / FOWEE#]	-----	-----
58 DUKKY	KFLL	SEAAN PEACH **OLAHS#**	-----	-----
	MYNN	SEAAN ZQA	-----	-----
	MYEF	DUKKY	-----	-----
MTEG	MTPP	-----	BOTES AOB FL260	-----
MDCS	MDPC	-----	BDRY AOB FL270	-----
	MDSD, MDJB, MDLR	-----	BDRY AOB FL230	-----
	MDPP, MDST, MDCY	-----	BDRY AOB FL150[11]	-----
SJU CTR	TJSJ	**CALTO Y290 BEANO** HARDE IDAHO RTE6 SJU	-----	-----
	TNCM	**CALTO Y290 SLUGO** HARDE IDAHO RTE6 SJU SLUGO	-----	-----

Table 48 - GTK Routes & Restrictions

FROM	TO	ROUTING	CONTROL GIVEN
----- NONE -----			

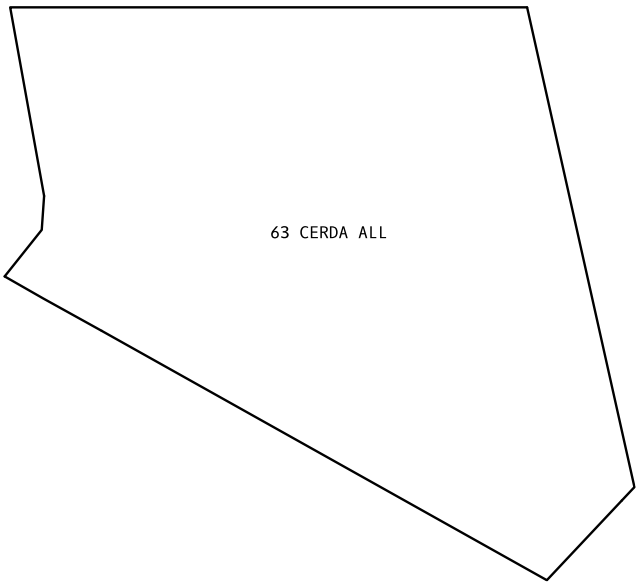
Table 49 - GTK Nonstandard Transfers of Control

[11] Note that this is genuinely "Flight Level 1-5-0", and not 15,000 feet, due to lack of nearby altimeters.

For VATSIM simulation purposes only. No affiliation with FAA.

Sector 63 CERDA

CERDA is the often-quiet northeastern sector of the Ocean area, bordering New York Oceanic. It is quite large, but the traffic is all predominantly right along its southern boundary, moving in the same southeast/northwesterly direction as the rest of the Ocean area.



Standard Callsign	ZMO_63_CTR
Alternate Callsign	ZMO_CE_CTR
Frequency	135.2
Altimeters	-----

Table 50 - CERDA Sector Details

SECTOR	DEST	ROUTING	ALTITUDE	SPEED
SJU CTR	TJSJ	**FIPEK Y355 PLING SAALR**	-----	-----
	TNCM	**FIPEK Y355 SLUGO**	-----	-----

Table 51 - CERDA Routes & Restrictions

FROM	TO	ROUTING	CONTROL GIVEN
----- NONE -----			

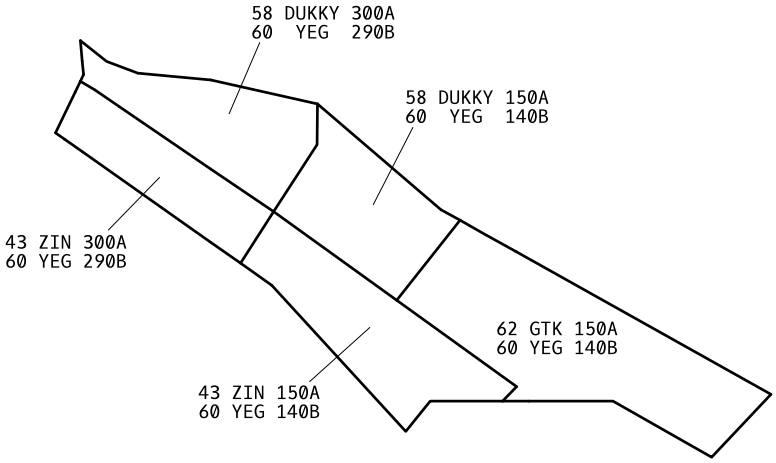
Table 52 - CERDA Nonstandard Transfers of Control

For VATSIM simulation purposes only. No affiliation with FAA.

Sector 60 YEG (Georgetown)

YEG is a large, low altitude sector, providing en route and approach control services to airports along the Bahamian island chains, plus Turks and Caicos.

When not open on its own, Provo Approach is covered top-down by YEG.[12]



Standard Callsign	ZMO_60_CTR
Alternate Callsign	ZMO_GE_CTR
Frequency	127.225
Altimeters	MYNN

Table 53 - YEG Sector Details

SECTOR	DEST	ROUTING	ALTITUDE	SPEED
MYNN	MYNN	SEAAN ZQA	SEAAN @ 12,000	-----
MBPV	MBPV	-----	DSDG TO 7,000	-----
41 VIICE	KMIA	MADIZ **VIICE#** MADIZ FOWEE FOWEE#	-----	-----
42 ZBV	KFLL	SEAAN PEACH **OLAHS#**	-----	-----
	KPBI	SEAAN PEACH **MAHHI#**	-----	-----

Table 54 - YEG Routes & Restrictions

FROM	TO	ROUTING	CONTROL GIVEN
58 DUKKY	60 YEG	ALL LDG TFC	TURNS & DESCENT
60 YEG	MBPV	ALL LDG TFC	TURNS

Table 55 - YEG Nonstandard Transfers of Control

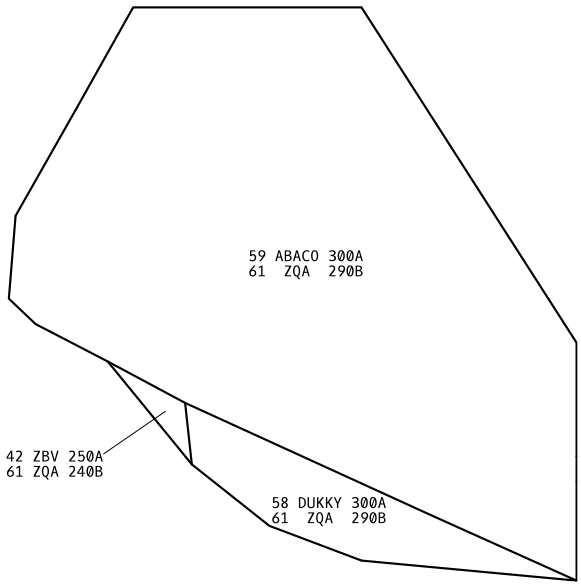
[12] As needed, Sector 60 may elect to delegate these approach controls to a different sector.

For VATSIM simulation purposes only. No affiliation with FAA.

Sector 61 ZQA (Nassau Low)

ZQA is a low altitude sector providing en route services to aircraft in the northern Bahamas, including the busy Nassau-Freeport corridor. ZQA works Nassau arrivals over HANKX and MAMML, Nassau departures via PEACH[13] and INGRA, plus various Freeport, Abaco, and overflight traffic.

When not open on their own, Nassau Approach and Freeport Approach are covered top-down by ZQA.[14]



Standard Callsign	ZMO_61_CTR
Alternate Callsign	ZMO_NA_CTR
Frequency	125.7
Altimeters	-----

Table 56 - ZQA Sector Details

SECTOR	DEST	ROUTING	ALTITUDE	SPEED
MYNNz	MYNN	HANKX	HANKX @ 11000	250K[15]
		MAMML	MAMML @ 12,000	-----
MYGFz	MYGF	-----	DSDG TO 7,000	-----
42 ZBV	KFLL (J)	SEAAN PEACH **OLAHS#**	---	-----
	KFLL (P)	SEAAN PEACH **KYAKS#**	PEACH AOB FL260	-----
	KPBI	SEAAN PEACH **MAHHI#**	---	-----
21 ZFP	KMIA	REPEE **BNFSH#**[16]	AOB FL240	-----
	KFLL	REPEE **CUUDA#**[16]	AOB FL200	-----
	KPBI	ZFP MAHHI **MAHHI#**	AOB FL200	-----

Table 57 - ZQA Routes & Restrictions

FROM	TO	ROUTING	CONTROL GIVEN
61 ZQA	MYGFz	ALL LDG TFC	TURNS

Table 58 - ZQA Nonstandard Transfers of Control

[13] While 61 does work departures via PEACH, note that departures via MAJUR will be handed off from Nassau Approach directly to Sector 42.
[14] As needed, Sector 61 may elect to delegate these approach controls to a different sector.
[15] Speed may be omitted if 15MIT or greater is already being provided.
[16] For Abaco departure traffic only. All other MIA/FLL arrivals should be rerouted by ZMO through 43/58/60.

For VATSIM simulation purposes only. No affiliation with FAA.

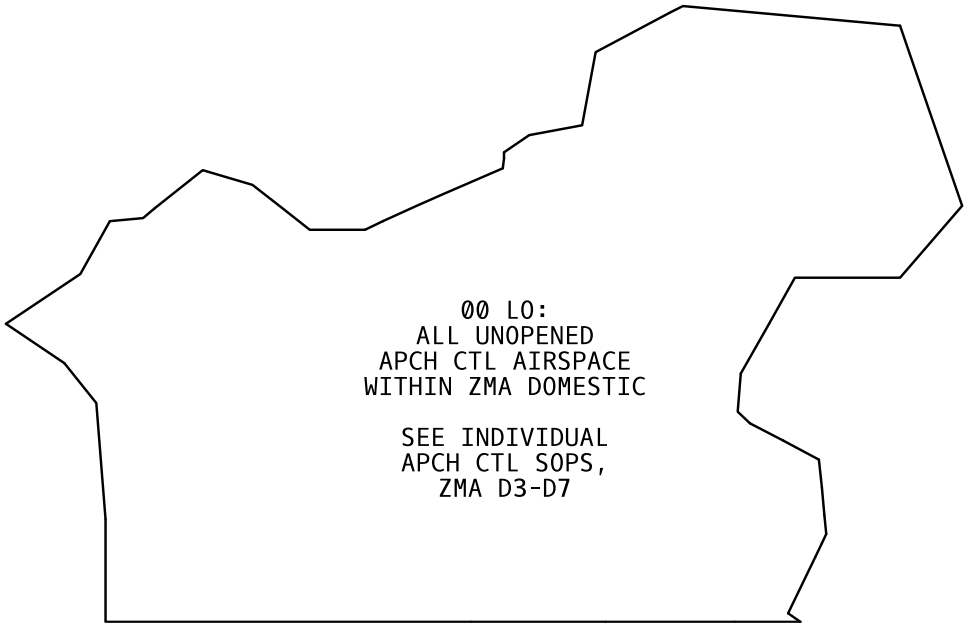
11. SPECIAL POSITIONS

Sector 00 LO (Low)

To assist with handling top-down demand, sector 00 can be opened to assume control of all approach control airspace within ZMA jurisdiction which has not been opened by an approach controller. This airspace DOES NOT include ANY airspace outside of TRACONs, and excludes TRACONs within ZMO[17]. For example, KSEF, KOBE or KMTH (and others) which are not contained within an approach control are worked by the appropriate center sector, not this sector. Likewise, low altitude or VFR flight following traffic outside of approach control airspace (such as through the middle of the state) shall not be worked by this position.

This position shall control all approach control airspace in accordance with all applicable directives pertaining to the approach control and tower facilities for which sector 00 is assuming responsibility.

This sector cannot be open alone, requiring at least one other center sector to be online.



Standard Callsign	MIA_LO_CTR
Alternate Callsign	MIA_UL_CTR
Frequency	132.25
Altimeters	Primary Fields of owned TRACONs

Table 59 - LO Sector Details

[17] Freeport, Nassau, and Provo approach controls are not included in MIA_LO_CTR.

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TMU: "Flow Control" – TMU Primary Position

The flow control position is the first TMU position to be opened. When multiple TMU positions are open, Flow Control is responsible for managing the overall flows through ZMA through interfacility coordination and implementation of the necessary Traffic Management Initiatives (TMIs). Acts as the primary point of contact for all coordination with external facilities.

TMU: "Departure Releases"

This position is responsible for issuing release times for all ZMA airports (or adjacent airports, as appropriate) which are currently under a Call-For-Release (CFR) TMI, as issued by the Flow Control position, or an external facility. This position will coordinate directly with the appropriate towers and maintain a list of the aircraft waiting to depart, including total delay time, and issue delay estimates if a release time is not immediately available/calculable.

TMU: "Routing"

This position is responsible for detecting bad routes for aircraft in neighboring or terminal facilities and taking the appropriate action to resolve those issues as early as possible. This position will also work closely with Flow Control in the event that sweeping reroutes become necessary because of weather considerations. Fields questions about potential route issues from internal and external facilities.

TMU Position	Network Callsign
Flow Control (default)	MIA_TM_CTR
Departure Releases	MIA_CF_CTR
Routing	MIA_RT_CTR

Table 60 – TMU Position Details

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